CLINICAL · MOUTH & THROAT

Benign oral lesions and premalignant patches

Lumps in the mouth are common and mostly benign: a fibrous polyp where the cheek is bitten, a pyogenic granuloma on the gum, a mucocele on the lower lip, a ranula under the tongue. Between these and cancer sits a group of patches, white and red, that are not yet malignant but may become so. Recognise each, excise what needs excising, and keep the premalignant patient under watch.

AT A GLANCE

- Leukoplakia is a clinical term: a white patch not otherwise explained.
- Erythroplakia carries far the higher malignant risk.
- Remove the ranula with its sublingual gland.
- Every palatal lump is biopsied: many minor salivary tumours are malignant.

1 CHAPTER ONE Overview

Three groups.

1.1 Classification

- **Benign tumours:** solid (papilloma, fibroma, haemangioma, lymphangioma, torus, pyogenic granuloma, granular cell tumour, minor salivary gland adenoma) and cystic (mucocele, ranula, dermoid).
- **Premalignant lesions and conditions:** leukoplakia, erythroplakia, submucous fibrosis and others.
- **Malignant lesions:** squamous cell carcinoma and the non-squamous cancers (see the oral and oropharyngeal cancer topic).

1.2 The rule

Any lump that is growing, ulcerated, indurated or fixed, or any patch that is red, speckled or nodular, is biopsied.

2 CHAPTER TWO Solid benign tumours

The common lumps.

2.1 Papilloma

Common, usually in the third to fifth decades, on the palate, uvula, tongue and lips: a small (under 1 cm), white, pedunculated growth with finger-like projections, linked to HPV 6 and 11. Excisional biopsy; recurrence is rare.

2.2 Fibroma (fibroepithelial polyp)

A smooth, mucosa-covered, pedunculated or sessile lump about 1 cm across, firm or soft, along the bite line of the cheek or on the tongue, caused by chronic irritation. Conservative excision.



FIGURE 1 A fibroepithelial polyp in the left cheek (arrow).
Dhingra, fig. 44.1

2.3 Haemangioma

Mostly in children, on the lip, tongue and cheek: a soft, red or blue, compressible lump that may bleed when bitten. Capillary, cavernous and mixed types are described. Lesions present at birth are observed, since many regress; propranolol and steroids help proliferating haemangiomas. Large, persistent or growing lesions are treated by embolisation (alone or before surgery), laser or excision;

sclerotherapy works poorly in diffuse lesions. In older adults, dilated veins (*phlebectasia*) can look similar.

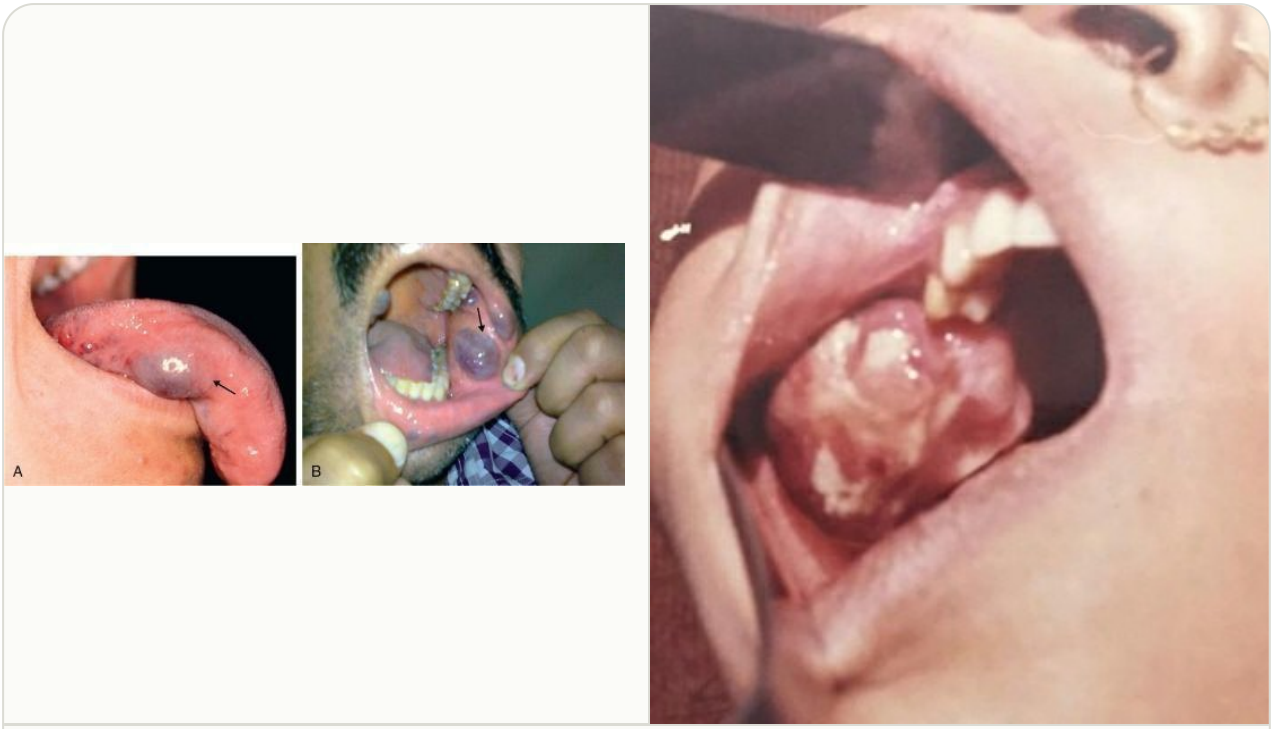


FIGURE 2 First, (A) a haemangioma on the lateral border of the tongue and (B) multiple haemangiomas of the lips, cheek and tongue in a man who bled when he bit them. Second, a haemangioma of the right cheek in a young girl.
Dhingra, figs 44.2 and 44.3

2.4 Lymphangioma

Usually in the anterior two-thirds of the tongue, as a localised compressible swelling or a diffuse enlargement causing *macroglossia*. It does not regress. Small lesions are excised; large symptomatic ones are debulked, since complete removal is rarely possible.

2.5 Torus

A bony outgrowth covered by mucosa: *torus palatinus* in the midline of the hard palate (the commoner), *torus mandibularis* on the lingual surface of the mandible near the premolars, usually on both sides. Harmless; removed only if it interferes with speech, chewing or dentures.

2.6 Pyogenic granuloma

A reactive overgrowth of granulation tissue after trauma or irritation, usually on the anterior gums, sometimes the tongue, lips or cheek: a soft, red to purple lump that bleeds on touch. Excision; recurrence is unlikely after complete removal.



FIGURE 3 A pyogenic granuloma.
Dhingra, fig. 44.4

2.6.1 Pregnancy granuloma

Identical clinically and histologically, arising in the first trimester (*epulis gravidarum*) and regressing after delivery; excised only if it persists, since it recurs if removed during pregnancy.

2.7 Granular cell tumour

A firm submucosal nodule, usually on the dorsum of the tongue, once called a myoblastoma but derived from Schwann cells; its histology shows pseudoepitheliomatous hyperplasia that can be mistaken for carcinoma. Excision. *Congenital epulis* is a related tumour on the gums of newborn girls.

2.8 Minor salivary gland tumours

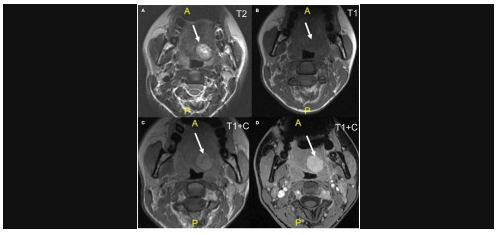
Pleomorphic adenoma is the commonest benign tumour, usually a painless submucosal lump on the hard or soft palate. Because a large proportion of minor salivary gland tumours are malignant, every palatal lump is biopsied, and adenomas are widely excised to prevent recurrence.

2.9 Other benign tumours

Neurofibroma and schwannoma (the tongue is a favoured site), lipoma (soft, yellow, in the cheek or floor), solitary fibrous tumour and verruciform xanthoma, all treated by excision.

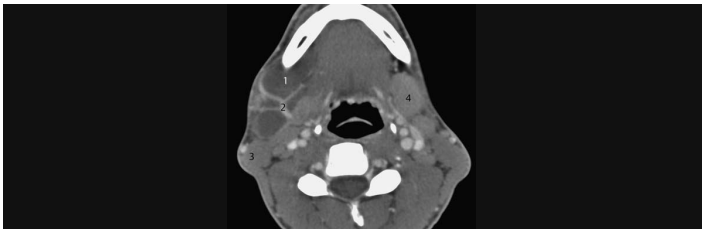
IMAGING IN THE APP 2 cases

Schwannoma of the oral cavity, and lymphangioma.



Schwannomas of the oral cavity and neck

After surgery · 3 images



Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

Congenital · 14 images

3 CHAPTER THREE Cysts

Mucocele, ranula, dermoid.

3.1 Mucocele

A retention or extravasation cyst of a minor salivary gland, almost always on the *lower lip*, after a bite: a soft, bluish, translucent swelling that may burst and refill. Excision with the underlying gland.



FIGURE 4 (A) A mucocele of the lower lip. (B) A mucocele on the ventral surface of the tongue tip.
Dhingra, fig. 44.5

3.2 Ranula

A cystic, translucent, bluish swelling in the floor of mouth to one side of the frenulum, pushing the tongue up, from obstruction of the ducts of the *sublingual gland*. A *plunging ranula* passes through or behind the mylohyoid into the neck and presents as a soft submandibular or submental swelling.



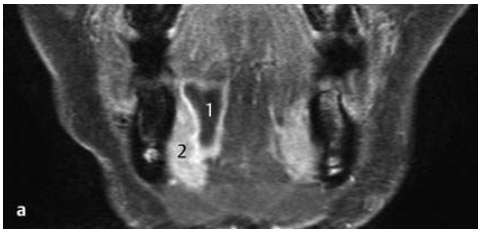
FIGURE 5 A ranula: a translucent swelling under the tongue.
Dhingra, fig. 44.6

3.2.1 Treatment

Dhingra describes excision for small ranulas and marsupialisation for large ones, noting that the thin wall makes complete removal difficult. Recurrence after marsupialisation alone is common; excision of the ranula *with the sublingual gland* through the mouth, protecting the submandibular duct and lingual nerve, is now preferred and also cures most plunging ranulas.

IMAGING IN THE APP 1 case

Ranula of the sublingual gland.



Ranula of the Sublingual Gland

Infection & inflammation · 10 images

OPERATIVE GUIDES IN THEATRE 1 operation

Ranula surgery.

Surgical Management of Ranula

Ranula and sublingual gland excision

3.3 Dermoid

A sublingual dermoid lies above the mylohyoid, in or beside the midline, and shines white through the mucosa, unlike the translucent ranula; a submental dermoid lies below the mylohyoid and bulges under the chin. Excised through the mouth or the neck according to its relation to the muscle.

SWELLINGS IN THE FLOOR OF MOUTH

FEATURE	RANULA	SUBLINGUAL DERMOID	SUBMANDIBULAR CALCULUS
Colour	Bluish, translucent	White, opaque	Normal mucosa over the duct
Consistency	Soft, fluctuant	Doughy	Hard, palpable along the duct

FEATURE	RANULA	SUBLINGUAL DERMOID	SUBMANDIBULAR CALCULUS
Position	One side of the frenulum	Midline or near it	Along Wharton's duct
Other	May plunge into the neck	Present from youth	Pain and swelling with meals

4 CHAPTER FOUR Premalignant lesions

White and red patches.

4.1 Definitions

A *premalignant lesion* is altered tissue in which cancer is more likely to develop (leukoplakia, erythroplakia); a *premalignant condition* is a general state with increased risk (submucous fibrosis, lichen planus, iron deficiency).

PRECANCEROUS LESIONS AND CONDITIONS OF THE ORAL CAVITY

- Leukoplakia and erythroplakia.
- Oral submucous fibrosis.
- Chronic hyperplastic candidiasis.
- Lichen planus (erosive and atrophic).
- Reverse-smoking palate and actinic cheilitis.
- Sideropenic anaemia (Plummer–Vinson syndrome).
- Pigmented lesions and melanosis; discoid lupus.

4.2 Leukoplakia

4.2.1 Definition

The WHO defines leukoplakia as a white patch that cannot be characterised clinically or pathologically as any other disease. It is a clinical term that says nothing about histology; lichen planus, candidiasis, leukoedema and white sponge naevus must be excluded.

4.2.2 Causes and sites

Smoking, tobacco and areca chewing, alcohol (especially with smoking), chronic trauma from teeth or dentures, candida, submucous fibrosis and Plummer–Vinson syndrome. In India the buccal mucosa and commissures are the commonest sites, at the place where the quid is held; the floor of mouth, tongue, gingivobuccal sulcus and lips are also affected. Mostly men in their forties.

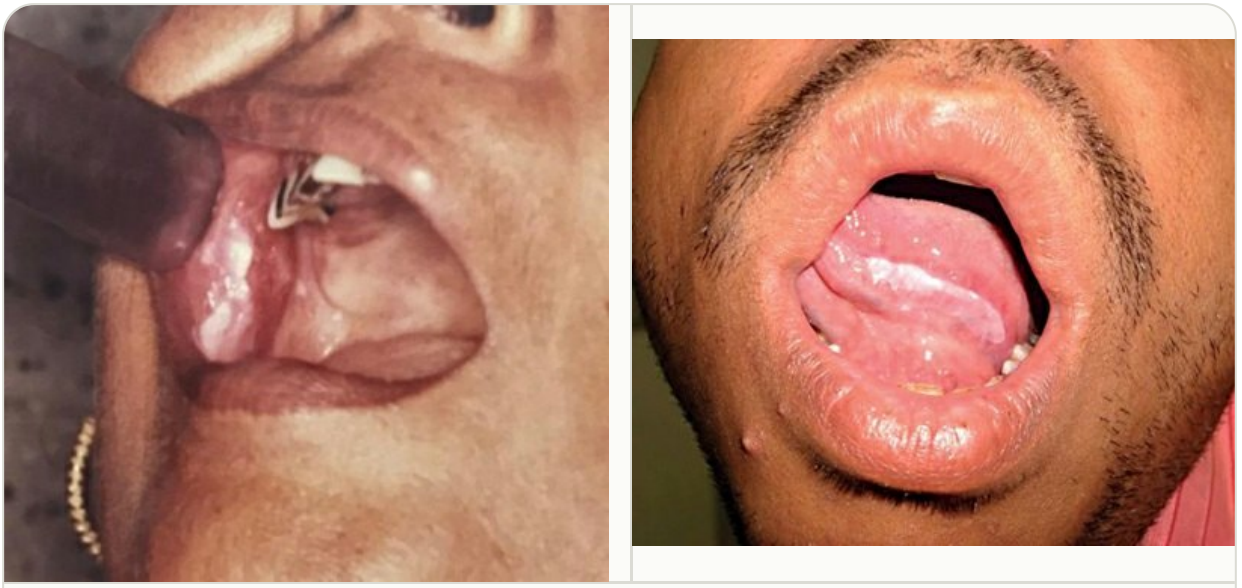


FIGURE 6 First, leukoplakia of the cheek after pan and tobacco use. Second, leukoplakia on the lateral border of the tongue.
Dhingra, figs 44.7 and 44.8

4.2.3 Types

- **Homogeneous:** a uniform, smooth or wrinkled white plaque; the lowest risk.
- **Speckled (nodular):** white nodules on a red base.
- **Erosive (erythroleukoplakia):** white mixed with red, with erosions and fissures.
- **Proliferative verrucous leukoplakia:** multifocal, warty and relentlessly progressive, often in older women without the usual risk factors; a very high risk of carcinoma.



FIGURE 7 Leukoplakia at the site where a quid of tobacco and lime was held.
Dhingra, fig. 44.9

4.2.4 Malignant potential

About a quarter show some dysplasia, and the higher the grade the greater the risk. Quoted rates of malignant change range from under 1% to 17.5%, averaging about 5%; speckled, erosive and verrucous forms, the floor of mouth and ventral tongue, and long follow-up carry the highest risk.

4.2.5 Management

- Stop tobacco, areca and alcohol, and remove sharp teeth: many lesions regress.
- Biopsy, from the reddest or most nodular area, to grade any dysplasia.
- Mild or no dysplasia: review regularly, and excise if it persists or changes.
- Moderate dysplasia or worse: excise, or ablate with laser or cryotherapy.
- Retinoids and beta-carotene can induce remission, but lesions tend to return when they are stopped.

4.3 Erythroplakia

A velvety red patch or plaque that cannot be characterised as any other disease, often slightly depressed with a sharp edge, on the floor of mouth, ventral tongue, soft palate and gingivobuccal sulcus. The redness is the vascular submucosa shining through thin, poorly keratinised epithelium. Most already show severe dysplasia, carcinoma in situ or invasive cancer when first seen, and the risk is many times that of leukoplakia. Excision biopsy and close follow-up.

RED IS WORSE THAN WHITE

Erythroplakia and the red parts of a speckled leukoplakia are where carcinoma is found. Biopsy the red.

4.4 Pigmented lesions

Benign melanotic macules and naevi are common; about a quarter of mucosal melanomas look benign at first, so a new, enlarging or irregular pigmented lesion on the palate or gum is biopsied.

4.5 Chemoprevention

Vitamin A, beta-carotene, vitamin E and retinoids such as 13-cis-retinoic acid can make leukoplakia regress in a quarter to a half of patients and may reduce second primary cancers, but the benefit lasts only as long as the drug is taken; stopping tobacco remains the most effective prevention.

5 CHAPTER FIVE Clinical comparison

Telling them apart.

ORAL WHITE PATCHES

LESION	WIPES OFF?	CLUE
Thrush	Yes, leaving a red base	Infant, antibiotics, steroids, diabetes
Leukoedema	Disappears on stretching	Bilateral buccal mucosa, a normal variant
Lichen planus	No	Lace-like Wickham striae, both cheeks
Hairy leukoplakia	No	Lateral tongue borders, HIV
Hyperplastic candidiasis	No	Commissures; responds partly to antifungals
Leukoplakia	No	Tobacco site; a diagnosis of exclusion

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definition of leukoplakia,** its types and malignant potential.
- **Erythroplakia** and why it is worse.
- **List of premalignant lesions and conditions.**
- **Ranula:** origin, the plunging type, and treatment by excision of the sublingual gland.
- **Mucocele, pyogenic granuloma** and **torus.**

6.2 Common pitfalls

COMMON PITFALLS

- Using "leukoplakia" as a histological diagnosis.
- Marsupialising a ranula and leaving the sublingual gland.
- Biopsying the white part of a speckled leukoplakia instead of the red.
- Excising a pregnancy granuloma during pregnancy.

Written for this app from Oxford Textbook of Otolaryngology ch. 57, Oxford Specialist Handbook ch. 6, Dhingra ch. 44 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

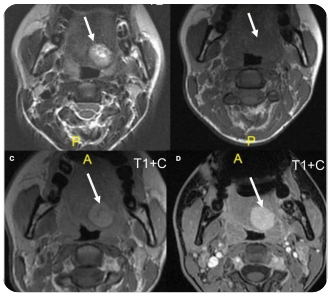
IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NECK: TRIANGLES, FASCIA & NODES · AFTER SURGERY

Schwannomas of the oral cavity and neck



Lingual nerve schwannoma (A (D) fat suppressed T1 post-contrast, demonstrating a lingual schwannoma (arrows) on the left side of the tongu…

SALIVARY GLANDS · INFECTION & INFLAMMATION

Ranula of the Sublingual Gland

IMAGING FINDINGS

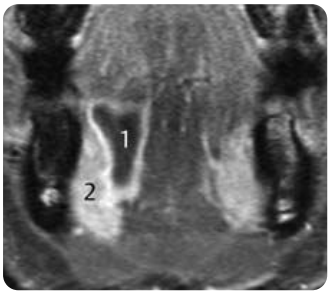
- Simple ranula: thin-walled cyst in the sublingual space
- Diving ranula: passes behind or through the mylohyoid into the submandibular space
- Tail sign pointing back to the sublingual space

TOP DIFFERENTIALS

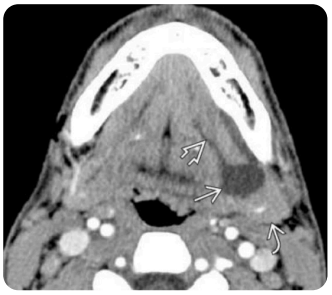
- Lymphatic malformation
- Dermoid or epidermoid
- Second branchial or thyroglossal cyst

CHECKLIST

- Relation to the mylohyoid decides the approach
- Remove the sublingual gland to prevent recurrence



MRI, T1-weighted with gadolinium enhancement and fat suppression, coronal. There is a cystic lesion suggestive of a ranula (1), originating…



Axial CECT demonstrates a rounded cystic mass in the submandibular space abutting the posterior margin of the mylohyoid muscle and displacing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

IMAGING FINDINGS

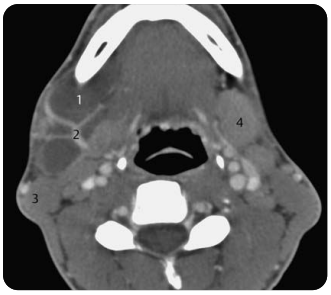
- Multicystic, septated, transspatial mass
- Fluid–fluid levels after bleeding
- Insinuates between structures without mass effect

TOP DIFFERENTIALS

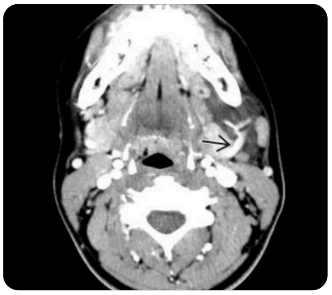
- Infantile haemangioma
- Venous malformation (phleboliths)
- Abscess

CHECKLIST

- Airway and mediastinal extent
- Macrocystic versus microcystic (sclerotherapy)



CT with intravenous contrast enhancement, axial. There is hypodense multilobular cystic structure (1) at the level of the…



Axial CECT in a 22-yearold woman with an angle of a mandible mass reveals a low-density submandibular space
Koch et al., Diagnostic Imaging: Head…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580.